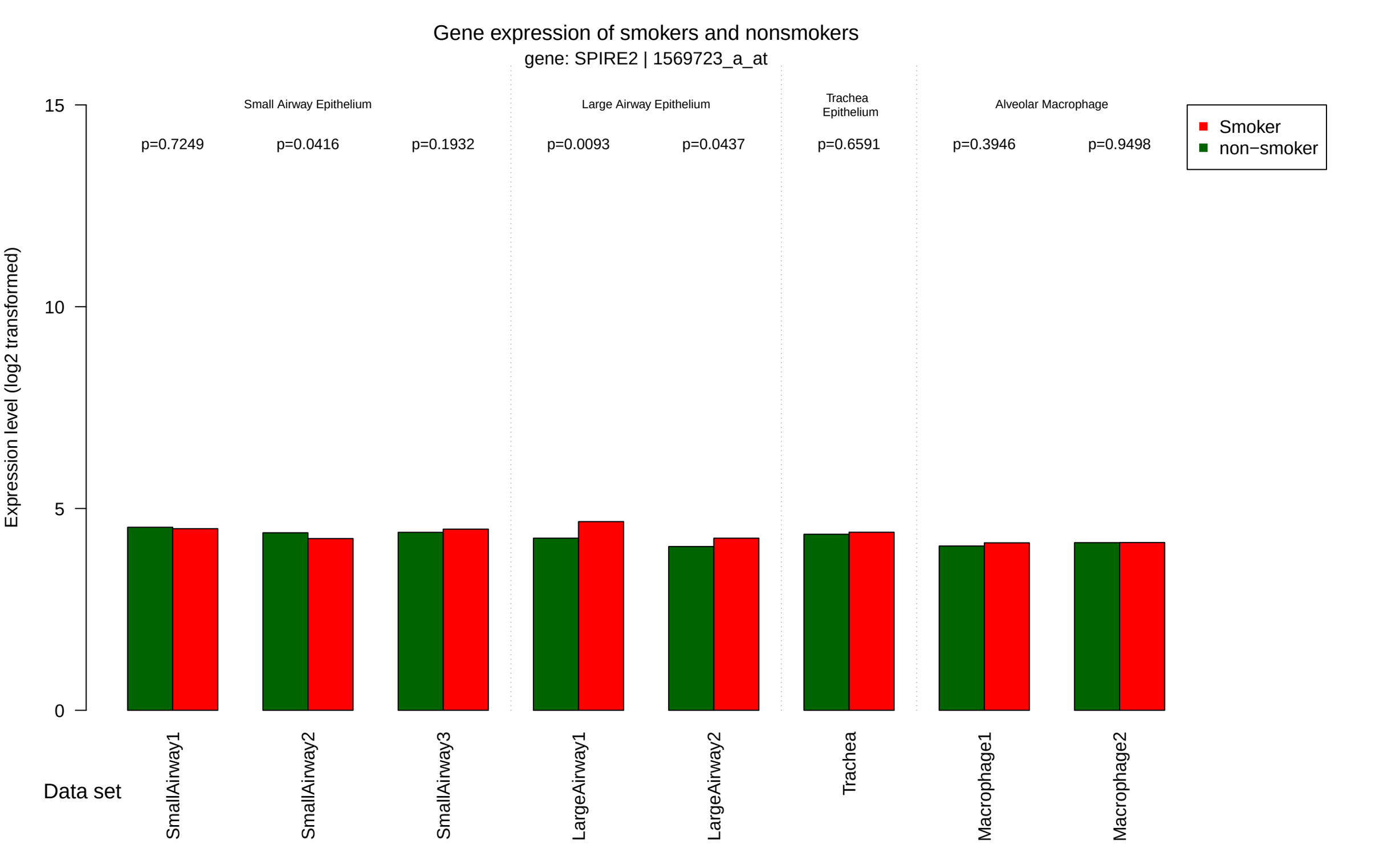
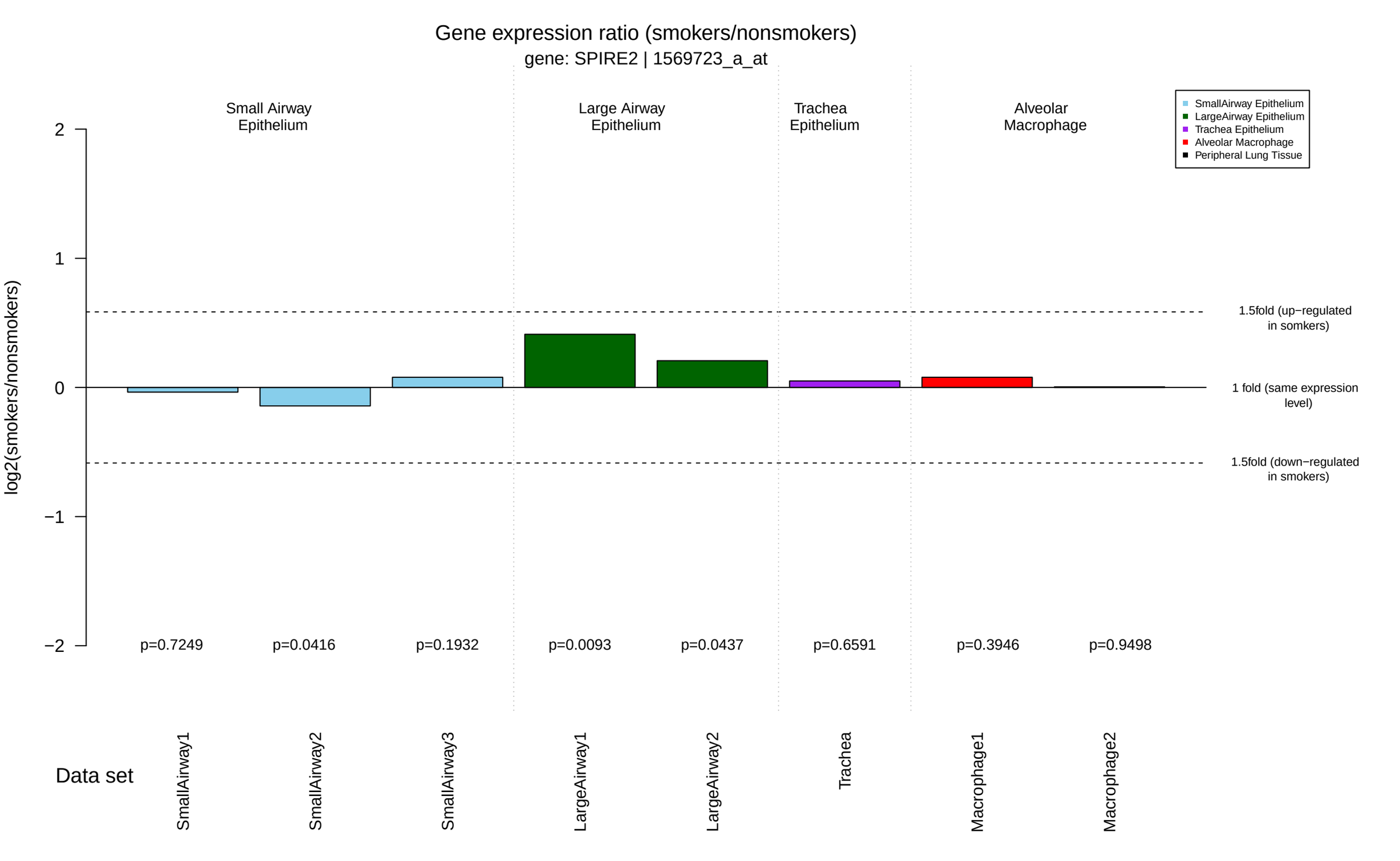
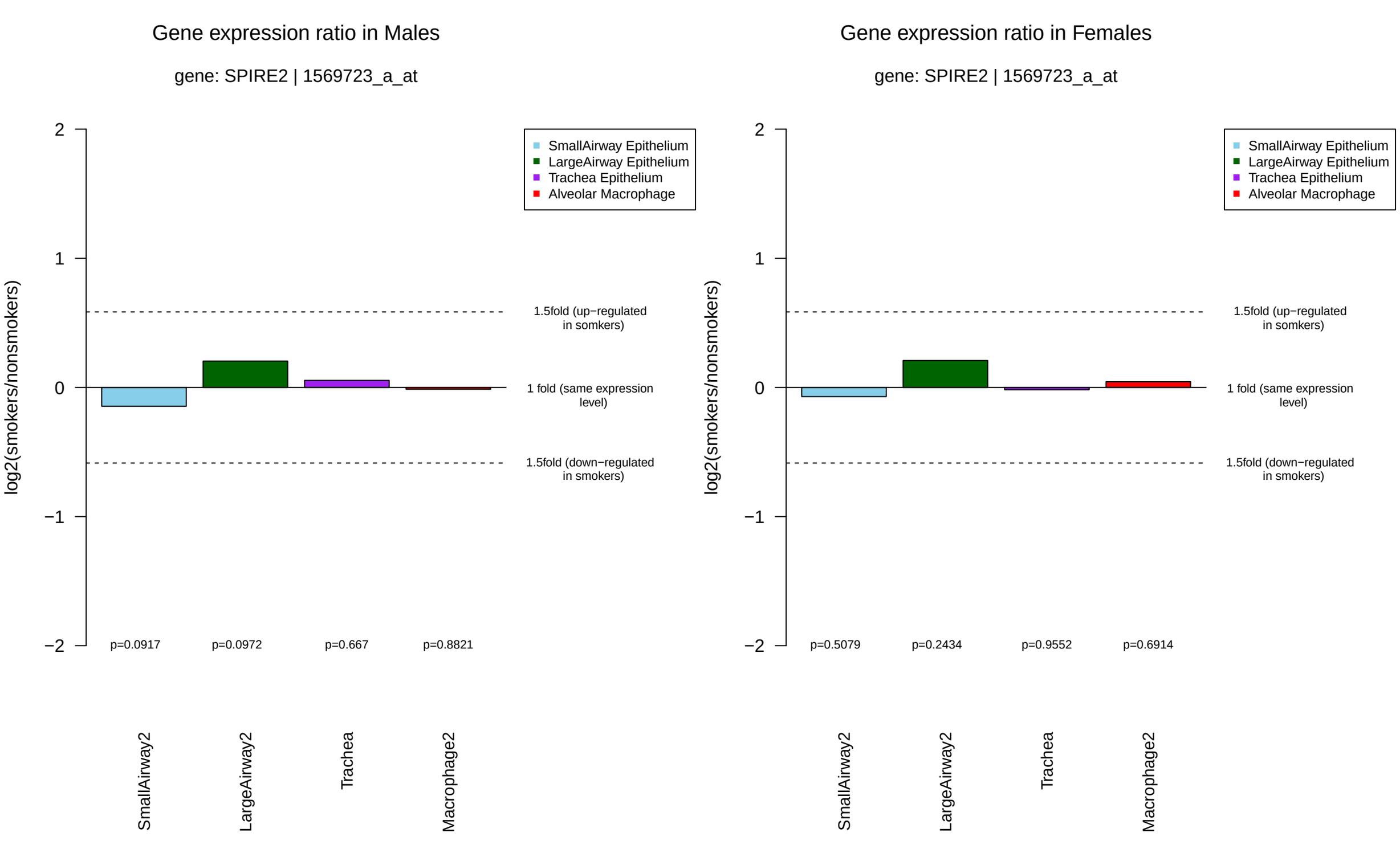
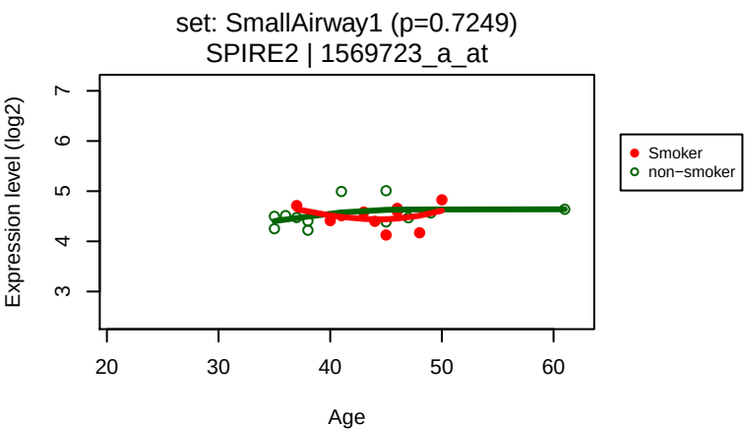
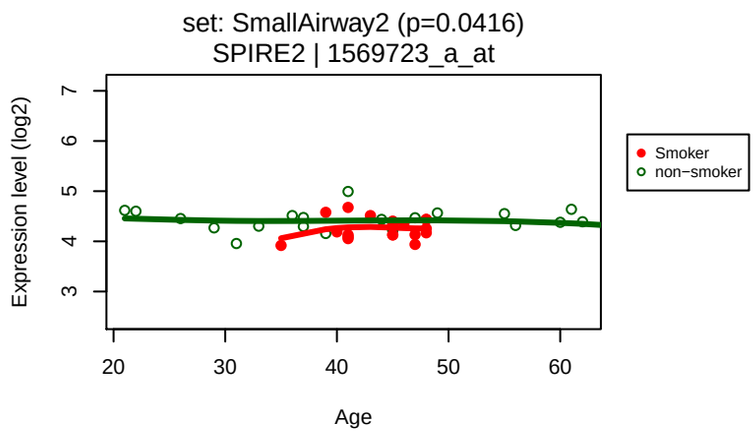
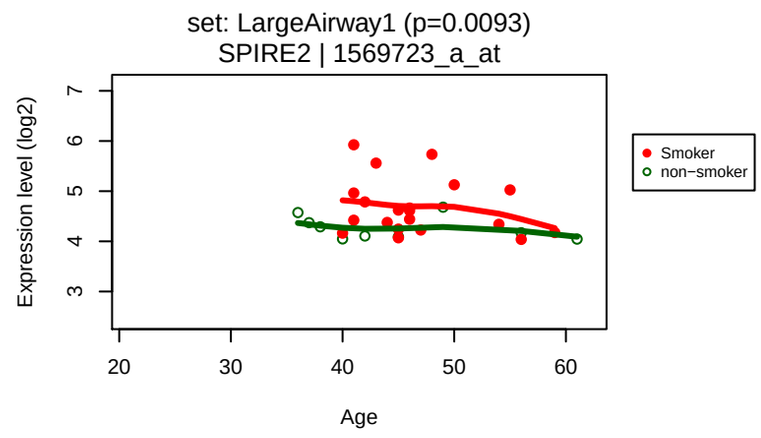

Gene expression of smokers and nonsmokers
gene: SPIRE2 | 1569723_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7249
p=0.0416
p=0.1932
p=0.0093
p=0.0437
p=0.6591
p=0.3946
p=0.9498
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

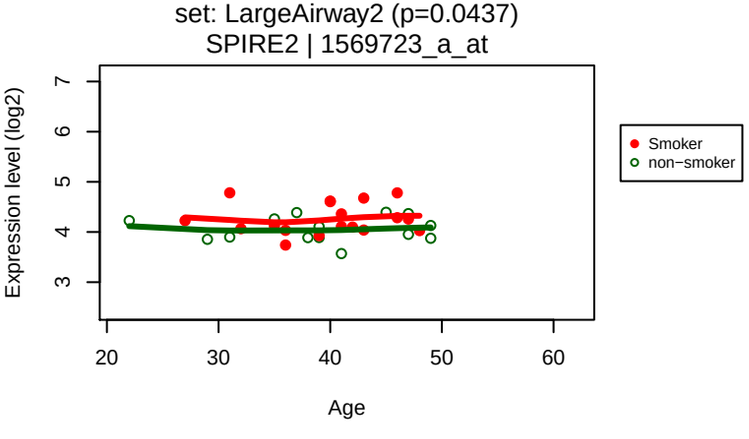

Gene expression ratio (smokers/nonsmokers)
gene: SPIRE2 | 1569723_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7249
p=0.0416
p=0.1932
p=0.0093
p=0.0437
p=0.6591
p=0.3946
p=0.9498
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

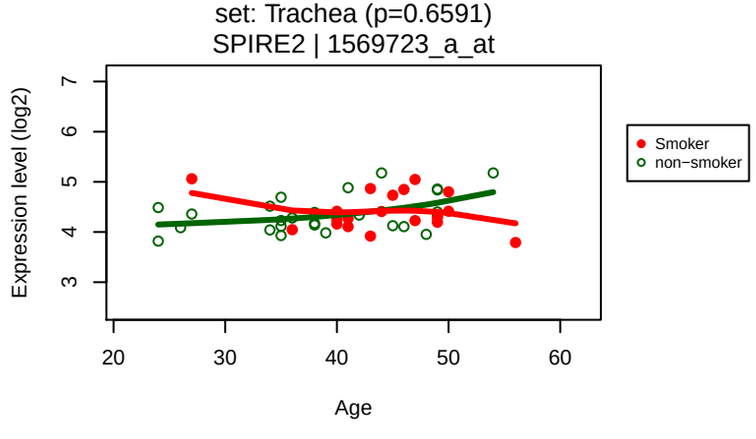

Gene expression ratio in Males
gene: SPIRE2 | 1569723_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0917
p=0.0972
p=0.667
p=0.8821
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPIRE2 | 1569723_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5079
p=0.2434
p=0.9552
p=0.6914
SmallAirway2
LargeAirway2
Trachea
Macrophage2

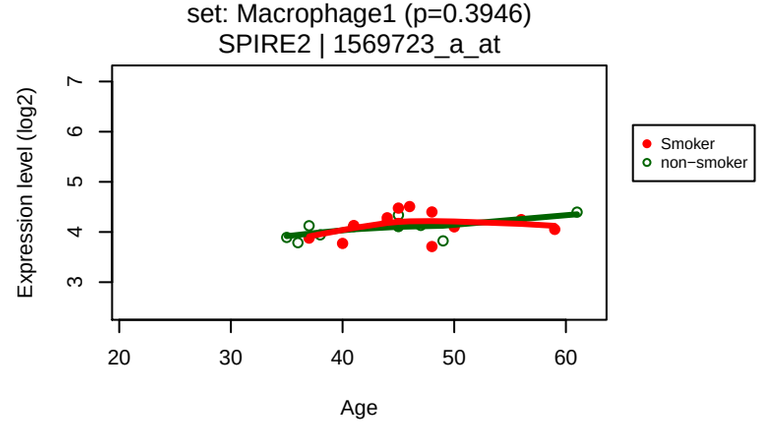

set: SmallAirway1 (p=0.7249)
SPIRE2 | 1569723_a_at
Expression level (log2)
Age
Smoker
non−smoker

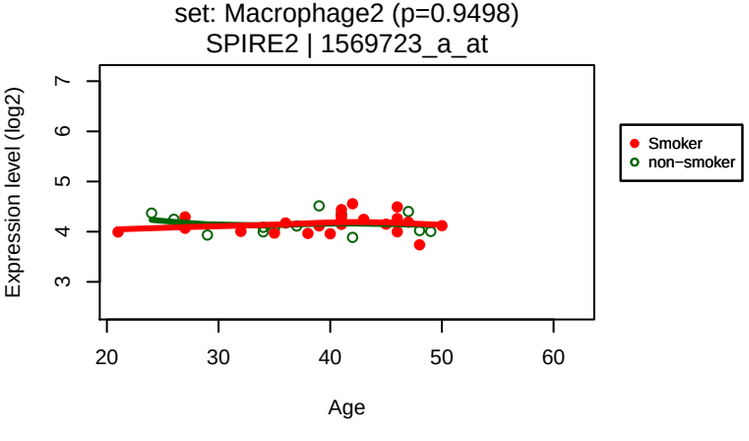

set: SmallAirway2 (p=0.0416)
SPIRE2 | 1569723_a_at
Expression level (log2)
Age
Smoker
non−smoker

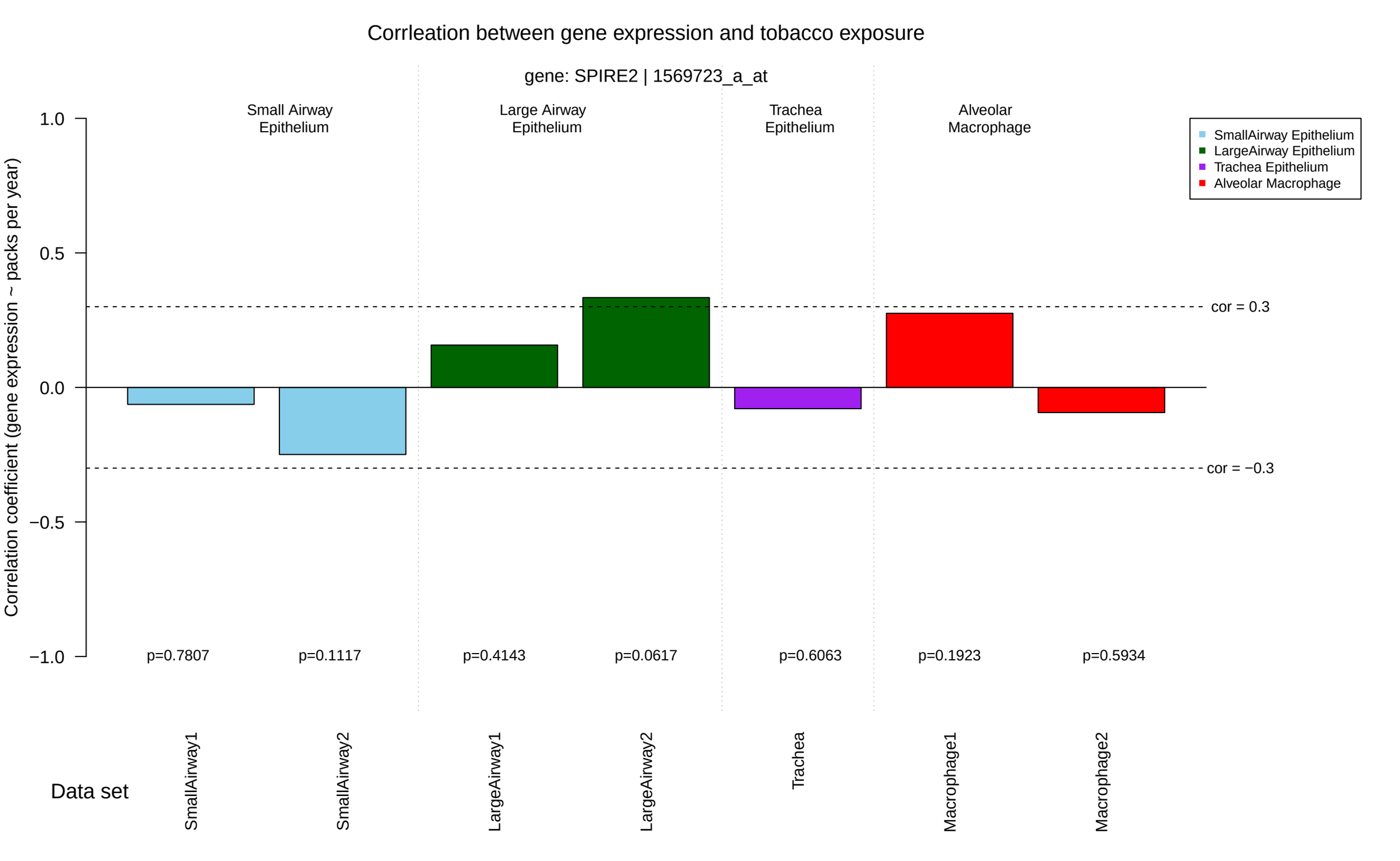

set: LargeAirway1 (p=0.0093)
SPIRE2 | 1569723_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0437)
SPIRE2 | 1569723_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6591)
SPIRE2 | 1569723_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3946)
SPIRE2 | 1569723_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9498)
SPIRE2 | 1569723_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPIRE2 | 1569723_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7807
p=0.1117
p=0.4143
p=0.0617
p=0.6063
p=0.1923
p=0.5934
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2